

U.S. Department of Commerce
National Institute of Standards and Technology
Material Measurement Laboratory
Chemical Sciences Division
Gaithersburg, MD 20899

REPORT OF ANALYSIS

07 August 2026

Submitted to: First Last, Title
Group or Division Name

Title: Title Line 1

Author(s): Brandon LeBlanc

SRM(s): NA

Assigned Value ($k = 2$): NA

Expiration Date: NA

Method(s): Sample preparation and/or instrumental methods used

Constituent(s): NA

Purpose: Brief statement specifying purpose of the ROA

Control(s): Control samples used

Date(s) of Lab Work: 01 January 9999 – 01 January 9999

Collaborator(s): NA

Quality Documents: Applicable quality manuals and technical procedures

Agreements: NA

Approvals: NA

Copies to: NA

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Physical Science Technician

Reviewed By:

Reviewer Name ✉
Reviewer Title

Introduction

This introduction is designed to give a brief overview of common markdown formatting tasks with Quarto. It is not designed to be exhaustive; the [Quarto guide](#) will be able to provide more examples and information.

italic text, **bold text**, ***bold italic text***, super^{script}, sub_{script}

Headings are shown throughout the document on the lines that begin with #. One # gives a level 1 heading, two #s give a level 2 heading, etc.

Quarto reserves certain symbols to specify markdown directives throughout the document, such as * and #. To insert these in normal text, they must be escaped (prepend a backslash).

- Bulleted lists
- must have a blank
- line separating them
 - from the surrounding text
 - or they won't
- render properly

For unordered lists, you can either use the symbols that the Quarto documents show, or you can just use a regular hyphen. Both will work, as long as the correct spacing is maintained.

Equations use LaTeX syntax. You can have equations as inline math: $E = mc^2$.

They can also be specified in their own block (display math):

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right] \Psi(x, t) \quad (1)$$

Equation 1 shows the Time-Dependent Schrödinger Equation, rendered in its own block.

Here's a second equation example:

$$a^2 + b^2 = c^2 \quad (2)$$

Equation 2 shows the Pythagorean Theorem.

Tables can be created in a variety of ways.

Table 1: Example of a pipe table.

header 1	header 2	header 3
row 1 col 1	row 1 col 2	row 1 col 3
row 2 col 1	row 2 col 2	row 2 col 3

Table 1 gives an example of a [pipe table](#), which is the most common way of specifying a markdown table. The number of hyphens used indicate the relative widths of the columns. The pipes do not have to be aligned with one another or spaced consistently. The cell contents of pipe tables can only be single-line elements, and keeping track of the formatting gets pretty cumbersome.

Table 2: Example of a list table.

header 1	header 2	header 3
row 1 col 1	row 1 col 2	row 1 col 3
row 2 col 1	row 2 col 2	row 2 col 3

Table 2 gives an example of a [list table](#). This format is much more readable and user-friendly for entering the data. Each top-level item indicates a new row, and each nested item is a cell in that row. The cell contents of list tables can be any content, including block elements (lists, code cells, etc.) and multiline paragraphs.

There are also [grid tables](#), which are similar to pipe tables but allow for arbitrary elements in the cells. Additionally, tables can be rendered programmatically using R or Python, which can be useful for reading in data from another file. This option requires using packages to output the data into a markdown format, and more details can be found [here](#).

Inserting figures is more straightforward, but there are a lot of layout customization options.

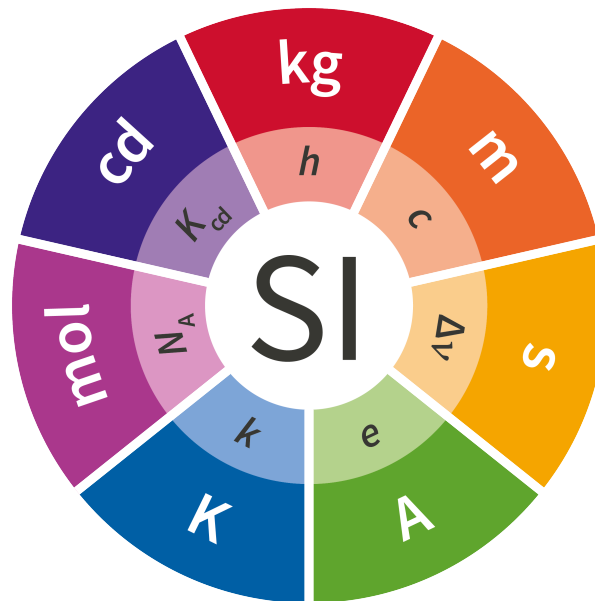


Figure 1: Defining the International System of Units (SI).

Figure 1 gives an example of inserting a figure with a caption. There are lots of ways to customize the layout of figures into multiple sub-figures, rows, columns, and panels. Like tables, figures can also be generated programmatically from R or Python, which is useful for including complex data visualizations in a report. Again, this requires the use of packages to generate the figures. More information on layouts and computed figures can be found [here](#).

Throughout this introduction, [cross-references](#) were used on equations, tables, and figures by assigning them a label beginning with # (e.g., #eq-tdse). These labels can be used in the body to reference the specific item by replacing the # with @. Equation labels must be prefixed with 'eq', table labels with 'tbl', and figure labels with 'fig'. Quarto will automatically keep track of the order they appear in the text and keep the references current should that order change.

TO DO:

- More examples for utilizing [references](#)
 - See the link to get started, and note that there is already a 'refs.yml' file in this template that can be updated with the relevant sources.
- Add examples of computations for tables and figures.
 - [R info](#)
 - [Python info](#)
- Add examples for [shortcodes](#), specifically for `include`, `pagebreak`, and possibly `contents`.

Traceability Statement

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Experimental

Safety

Hazard Reviews.

Materials

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Sample Preparation

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Equipment and Instrumentation

- Instrument/Equipment 1
- Instrument/Equipment 2
- Instrument/Equipment 3

Instrumental Methods

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Quantitative Methods

Statistical analysis

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Homogeneity analysis

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Stability

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Results and Discussion

Value Assignment of Lot Standards

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Analysis of previous lot standards

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Value Assignment of Samples

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Uncertainty Assessment

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Disposition of Cylinders

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Data Storage

- Notebook 1
- File 1
- File 2

References